Experiments
Reproduction of High-Dimensional MSWD Testing

MSWD Bootstrapping Reproduction

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How the paper connects the experiments

- Goal: test $H_0 : P = Q$ versus $H_1 : P \neq Q$ in high dimension.
- Core statistic: maximize one-dimensional Wasserstein distance over sparse directions.

$$T_n = \left(\frac{n_1 n_2}{n_1 + n_2}\right)^{1/2} \sup_{v \in V_{0,s}} W_1(v^\top X, v^\top Y).$$

- Decision: reject if observed T_n exceeds the bootstrap $1 - \alpha$ quantile.
- SCI: the same bootstrap construction also localizes marginal genes and GO-term subsets.

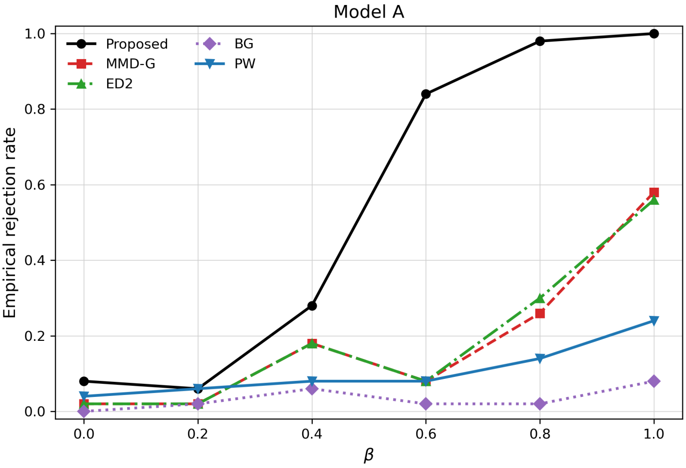
Three experiments and their roles

Experiment	Question	Output
Model A comparison	Does MSWD have a power advantage in the core simulation?	Empirical rejection curves from 50 runs
MSWD-only bootstrap	How does the statistic compare with its null distribution?	Observed T_n, threshold, p-value
GBM real data	Are LGG and GBM methylation distributions different, and where?	Global test, 138 genes, 7 GO terms

Experiment 1 setup: Model A, first 50 Monte Carlo runs

- Model: $X \sim N(0_p, \Sigma)$, $Y \sim N(\mu, \Sigma)$.
- Covariance: $\Sigma_{ij} = 0.5^{|i-j|}$.
- Decaying mean shift: $\mu_j = 0.8\beta j^{-3}$.
- Parameters: $n_1 = n_2 = 250$, $p = 500$, $\alpha = 0.05$.
- Signal grid: $\beta = 0, 0.2, 0.4, 0.6, 0.8, 1.0$.
- Reported as 50 complete runs per β; Proposed bootstrap $B = 300$.
- Comparators: MMD-G, ED_2, BG, PW; MMD-L is not recoverable from logs.

Experiment 1 result: Proposed gains power fastest



Experiment 1 interpretation

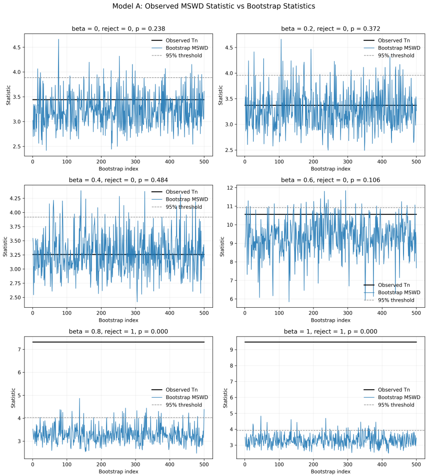
β	Proposed	MMD-G	ED_2	BG	PW
0.0	0.08	0.02	0.02	0.00	0.04
0.4	0.28	0.18	0.18	0.06	0.08
0.6	0.84	0.08	0.08	0.02	0.08
0.8	0.98	0.26	0.30	0.02	0.14
1.0	1.00	0.58	0.56	0.08	0.24

- At $\beta = 0$, size is about 0.08 for Proposed, plausible with 50 runs.
- From $\beta = 0.6$, Proposed separates sharply, matching the paper's message on sparse decaying signals.

Experiment 2 setup: MSWD-only bootstrap

- One Model A data set is generated for each β.
- Each data set receives $B = 500$ bootstrap replications.
- This is not a power experiment; it visualizes the test decision.
- Decision rule: observed T_n greater than the 95% bootstrap threshold.
- Parameters: $n_1 = n_2 = 250$, $p = 500$, $\alpha = 0.05$.

Experiment 2 result: strong signals exceed the threshold



Experiment 2 interpretation

β	observed T_n	threshold	p-value	reject
0.0	3.442	3.885	0.238	no
0.2	3.372	3.958	0.372	no
0.4	3.256	3.918	0.484	no
0.6	10.557	10.920	0.106	no
0.8	7.319	4.025	0.000	yes
1.0	9.479	3.921	0.000	yes

- $\beta = 0.6$ is a borderline non-rejection in this single draw.
- A zero empirical p-value with 500 bootstraps should be reported as $p < 0.002$.

Experiment 3 setup: LGG/GBM DNA methylation

- Data: TCGA LGG and GBM DNA methylation.
- Samples: $n_1 = 511$ LGG and $n_2 = 150$ GBM.
- Features: $p = 207$ glioma prognosis-related genes.
- Method: L0 sparse MSWD + bootstrap, $B = 500$, $\alpha = 0.05$.
- Aim: test global distributional equality and localize the differences.

Experiment 3 global result: LGG differs from GBM

Metric	Value
Test statistic	42.9236
Bootstrap critical value	1.8479
Empirical p-value	$p < 0.002$
Significant marginal genes	138 / 207
Significant GO terms	7 / 7

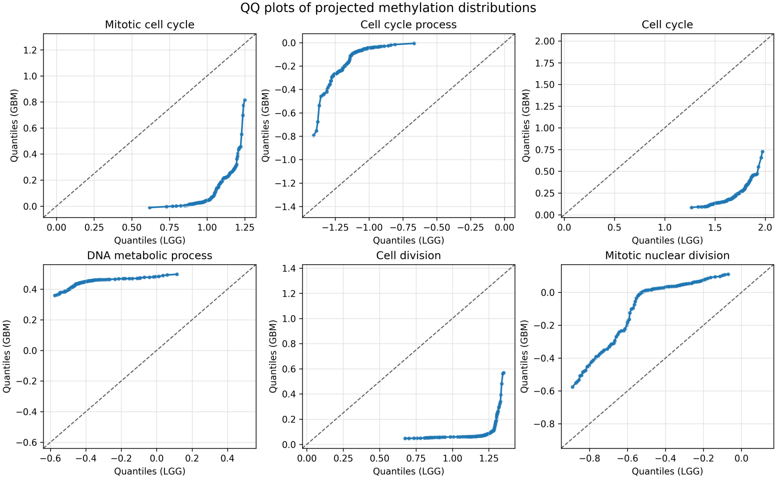
- The global statistic is far above the critical value.
- The paper reports 139 significant marginal genes; this reproduction finds 138.

Experiment 3 GO-term result

GO term	Genes	Statistic
mitotic cell cycle	5	9.467
mitotic cell cycle process	4	9.465
cell cycle process	7	10.196
cell cycle	11	15.454
DNA metabolic process	6	8.730
cell division	5	11.126
mitotic nuclear division	3	4.131

All seven subsets are significant, concentrating the difference in cell-cycle, DNA-metabolic, cell-division, and mitotic processes.

Experiment 3 QQ plots: projected distributions depart visibly



Integrated conclusion

- Experiment 1: MSWD has the strongest power under sparse decaying mean shifts.
- Experiment 2: bootstrap thresholds explain the Proposed method's rejection decision.
- Experiment 3: real GBM data show significant differences globally, marginally, and at the GO-term level.
- Methodological reason: max-sliced search reduces signal dilution, and SCI unifies testing with localization.